

Q #01

We have

$$D_1 = \frac{1}{2} D_2$$

Also, $P_1 = P_2$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\therefore A = \pi D^2 / 4$$

This becomes

$$\frac{F_1}{(\pi/4) D_1^2} = \frac{F_2}{(\pi/4) D_2^2}$$

$$\frac{F_1}{(\pi/4) (1/2 D_2)^2} = \frac{F_2}{(\pi/4) D_2^2}$$

$$\frac{F_1}{(\pi/4) 1/4 D_2^2} = \frac{F_2}{\pi/4 D_2^2}$$

$$\frac{F_1}{16} = \frac{F_2}{4}$$

$$\frac{\pi D_2^2}{4} F_1 = \frac{\pi D_2^2}{16} F_2$$

$$F_1 = \frac{F_2}{4} \text{ also, } F_2 = 4F_1$$

$$S = \frac{W_2}{F_2}$$

$$S = \frac{25}{300}$$

$$S = 0.0833 \text{ m} \rightarrow \text{For slave cylinder}$$

Now, work output of the slave cylinder

$$W = F \times S$$

$$W = (300)(0.0833)$$

$$W = 24.99 \text{ J} \approx 25 \text{ J}$$

Q #4

$$P_1 = 1000 \text{ kPa} \quad T_1 = 140^\circ\text{C}$$

$$m = 5 \text{ kg} \quad x = 25\% = 0.25$$

$$W = ?$$

For P_1 & T_1 we have from Table (B.5.2)

$$v = 0.83150 \text{ m}^3/\text{kg}$$

As piston's are connected in line

$$P_1 = P_2 = 1000 \text{ kPa}$$

Now, by doing interpolation, (Table B.5.2)

$$P = 1000$$

$$v = 0.83150$$

$$v = 0.83150$$

$$v_1 = m \times v = (1.5)(0.83150) = 1.24725 \text{ m}^3$$

$$v_2 = m \times v = (1.5)(0.47424) = 0.71136 \text{ m}^3$$

$$\text{Work} = P_{\text{avg}} \times \Delta v$$

$$= \left(\frac{P_1 + P_2}{2} \right) (v_2 - v_1)$$

$$\text{Work} = \left(\frac{600 + 200}{2} \right) (0.71136 - 1.24725)$$

$$\text{Work} = -291.24 \text{ kJ} \quad (\text{negative sign work is done})$$

$$\text{For Heat Transfer}$$

$$Q = du + P dv$$

$$Q = (u_2 - u_1) + W$$

$$u_2 = 2881.12 \text{ kJ/kg} \quad u_1 = 2576.87 \text{ kJ/kg}$$

$$u_2 - u_1 = 456.63$$

$$Q = 456.63 + (-291.24)$$

$$Q = 165.135 \text{ kJ}$$

$$P_2$$

$$P_1$$

$$v$$

$$v$$

$$v$$

$$v$$

$$v$$

$$v$$

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4. A cylinder with a frictionless piston contains 5 kg of superheated R-134a vapor at an initial state of 1000 kPa and 140°C. The system undergoes cooling at constant pressure until the refrigerant reaches a quality of 25%. Determine the work done during this process. [10 Marks]

5. A piston-cylinder setup holds 1.5 kg of water at an initial condition of 200 kPa and 150°C. The system undergoes heating, following a process in which pressure changes linearly with volume, reaching a final state of 600 kPa and 350°C. Determine the final volume, the work done, and the heat transfer during this process. Also draw a P-v diagram. [10 Marks]

6. A cylinder with a piston restrained by a linear spring contains 2 kg of carbon dioxide at 500 kPa, 400°C. It is cooled to 40°C, at which point the pressure is 300 kPa. Calculate the heat transfer for the process. [10 Marks]

Q #2

$$m = 17,500 \text{ kg}$$

$$v = 30 \text{ m/s}$$

$$P = 1250 \text{ kPa}$$

$$K.E. = \frac{1}{2} m v^2$$

$$K.E. = \frac{1}{2} (17500)(30)^2$$

$$K.E. = 7875000 \text{ J}$$

As, it contributes 30% of energy

$$E_{\text{piston}} = (7875000) \left(\frac{30}{100} \right)$$

$$E_{\text{piston}} = 2362500 \text{ J}$$

Now,

$$P_{\text{avg}} = \frac{P_1 + P_2}{2} = \frac{(1000) + (100)}{2}$$

$$P_{\text{avg}} = 550 \text{ kPa}$$

We know

$$W = P \Delta V$$

$$W_{\text{piston}} = P_{\text{avg}} \cdot V_{\text{displacement}}$$

$$V_{\text{displacement}} = \frac{W_{\text{piston}}}{P_{\text{avg}}}$$

$$V_{\text{displacement}} = \frac{2362500}{550 \times 10^3}$$

$$V_{\text{displacement}} = 4.295 \text{ m}^3$$

$$v = v_f + x v_{fg}$$

$$v = (0.0008735) + (0.25)(0.019172)$$

$$v = 0.00565 \text{ m}^3/\text{kg}$$

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7. An adiabatic steam turbine in a power plant processes 5 kg/s of steam entering at 3000 kPa and 500°C. During operation, 20% of the steam is extracted at 1000 kPa and 350°C to supply a feedwater heater, while the remaining steam exits at 200 kPa and 200°C. Calculate the power output of the turbine. [10 Marks]

8. In a control volume, two steady flows of air enter while one exits. The first inlet stream has a flow rate of 0.025 kg/s at 350 kPa and 150°C (state 1). The second inlet stream enters at 450 kPa and 15°C (state 2). A single outlet stream exits at 100 kPa and -40°C (state 3). The control volume rejects 1 kW of heat to the surroundings and produces 4 kW of power. Neglecting kinetic energy effects, determine the mass flow rate of the air entering at state 2. [10 Marks]

Q #3

$$A_1 = 5 \text{ cm}^2$$

$$P_1 = 1000 \text{ kPa}$$

$$W_1 = 25 \text{ J}$$

$$A_2 = 3 \text{ cm}^2$$

$$F_2 = ?$$

$$S = ?$$

$$F = (P)(A)$$

$$F = (1000 \times 10^3) (5 \times 10^{-4})$$

$$F = 500 \text{ N}$$

$$F = 500 \text{ N} \rightarrow \text{Master cylinder}$$

$$\text{Now, again for Slave cylinder}$$

$$F = (P)(A)$$

$$F = (1000 \times 10^3) (3 \times 10^{-4})$$

$$F = 300 \text{ N}$$

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